

Nordfels — level handbook

Generated from the shipped asset files (build index 5, scene file `LevelL5`). Every number is decoded or computed from them; formulas are stated where used. Provenance and checks: verification.

1. At a glance

Item	Value
Unlock requirement	beat Neuland(Tutorial)
Nights (waves)	13
Enemies in total	576
Total enemy base HP	17735
Gold coins dropped by waves (sum of <code>goldCoins</code>)	88
Flying enemies / bosses	70 / 0
Enemy spawn lines	7
Build slots	69
Shrines	6
Starting gold (<code>goldBalanceAtStart</code>)	8
Wave generator asset	Season2 Wave Gen
Perk slots (<code>maxPerkCount</code>)	5
Fixed loadout	none
Castle position (x, y, z)	-4.5, 0, 1.1
Hold-to-call-night time (<code>nightCallTime</code>)	2.0 s (key: Space)
Hero speed (walk / day-walk / sprint / day-sprint)	14.0 / 18.2 / 23.0 / 29.9 m/s
Interaction / coin-magnet radius	3.0 m / 7.0 m

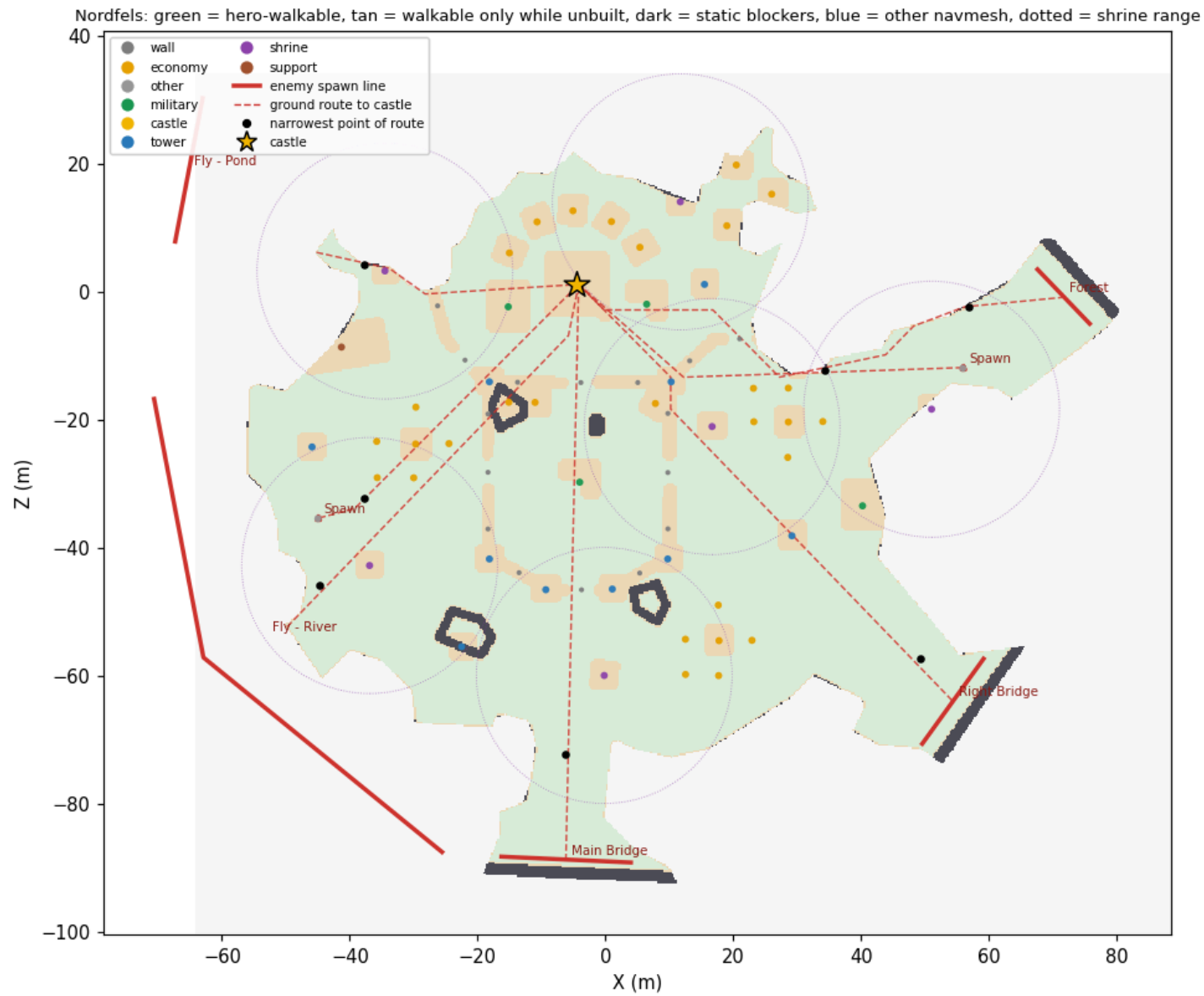


Figure 1. Hero-walkable area (green), blockers, build slots by type, enemy spawn lines (red), least-cost ground routes to the castle and their narrowest points. Built from colliders + the baked A navmesh; see §7.*

2. Where enemies come from

Enemies spawn at random points along a **spawn line** (a polyline of child transforms, `Spawn.cs:274-297`). The route is the least-cost path on the baked 'Enemy Units S' navmesh from the line's centre to the castle; travel time = route length ÷ enemy speed.

Spawn line	Centre (x, z)	Line length m	Can spawn	Difficulty	Route to castle m	Narrowest clearance m	Choke (x, z)	Shares path with
Right Bridge	(54, -64)	16.4	air/small/big	Normal	89	5.2	(49, -57)	Forest
Main Bridge	(-6, -89)	20.3	air/small/big	Normal	91	5	(-6, -72)	–
Fly - River	(-53, -54)	89.2	air	Normal	72	2.5	(-45, -46)	Fly - River
Spawn	(56, -12)	1.6	air/small/big	Normal	67	0.5	(34, -12)	–
Spawn	(-45, -35)	1.6	air/small/big	Normal	56	14.3	(-38, -32)	–
Forest	(72, -1)	11.8	air/small/big	Normal	87	0.5	(57, -2)	Right Bridge
Fly - Pond	(-65, 19)	22.8	air	Normal	44	0.5	(-38, 4)	Fly - Pond

3. Night by night

`EnemySpawner` mode: **Classic**; wave generator asset: `Season2 Wave Gen`; `pauseSpawningAtEnemyCount` = 275 (spawning pauses while that many enemies are alive; see waves doc). Times are seconds after dusk.

Night	In-game warning	Enemies	Base HP	Ranged	Flying	Boss	Coins	Composition	By spawn line	First contact s	Last arrival s
1	3 Swordsmen approaching from southern bridge	3	75	0	0	0	5	3× Swordsman	Main Bridge:3	18	24
2	8 Swordsmen approaching from the forest	8	200	0	0	0	5	8× Swordsman	Forest:8	17	24
3	2 Hunterlings coming from the forest will hunt you, 8 Melee coming from the southern bridg	10	260	0	0	0	5	8× Swordsman, 2× Hunterling	Forest:2, Main Bridge:8	9	22
4	24 Archers from all directions	24	480	24	0	0	8	24× Archer	Forest:8, Right Bridge:8, Main Bridge:8	18	41
5	15 Racers from the southern bridge will try to run straight to your castle center to destr	15	525	0	0	0	2	15× Racer	Main Bridge:15	9	23
6	20 Swordsmen and 20 archers arriving from the southern bridge	40	900	20	0	0	5	20× Swordsman, 20× Archer	Main Bridge:40	18	47
7	15 Flyers coming over the pond in the west	15	375	15	15	0	5	15× Wasp	Fly - Pond:15	8	22
8	30 Melee from the southern bridge, 10 Hunterlings and 10 Racers coming from the forest	50	1400	0	0	0	9	30× Swordsman, 10× Racer, 10× Hunterling	Main Bridge:30, Forest:20	9	47
9	5 Ogres coming from the southern bridge, 25 Flyers trickling in from all directions	30	1625	25	25	0	13	25× Wasp, 5× Ogre	Main Bridge:5, Fly - River:10, Right Bridge:5, Forest:5, Fly - Pond:5	8	47
10	90 Melee Units and 3 Oggers from all directions	93	2850	0	0	0	12	90× Swordsman, 3× Ogre	Forest:31, Right Bridge:31, Main Bridge:31	17	62
11	30 Racers & 50 Archers from southern bridge	80	2050	50	0	0	14	50× Archer, 30× Racer	Main Bridge:80	9	43
12	There is no serious attack tonight. Only a few scouts. The enemy is preparing for their fi	5	150	0	0	0	5	5× Hunterling	Right Bridge:5	9	13
13	Final attack. Catapults, ogres, archers, flyers. Make Your Last Stand! It's gonna be a lon	203	6845	76	30	0	0	80× Swordsman, 35× Racer, 30× Wasp, 30× Archer, 16× Catapult, 7× Ogre, 5× Hunterling	Main Bridge:86, Fly - River:15, Fly - Pond:5, Right Bridge:55, Forest:42	9	166

First contact = min over spawn groups of (`deLay` + route ÷ speed); last arrival = max of (`deLay` + (count–1)·`interval` + route ÷ speed). Flyers use the straight-line distance to the castle. These are lower bounds: enemies stop to fight units/buildings that they target on the way (see combat doc).

Derived: the heaviest night by total base HP is night 13 (6845 HP, 203 enemies); night-to-night HP ratio (last/first) = 91.3×. Ranged enemies appear on nights 4, 6, 7, 9, 11, 13; flyers on nights 7, 9, 13.

4. Enemy roster

Name	Prefab	HP	Speed	Range	Damage	Cooldown	DPS	Tags	In-game description
Archer	E Archer	20	5	20	2.5	2	1.25	RangedFighter, TakesIncreasedDamageFromTowers, NaturallyVulnerableToSplash, Humanoid	Archers shoot arrows from afar. Quite strong when left uncontested. Weak against buildings and towers.
Swordsman	E Melee	25	5	3	2.5	1	2.5	MeeleFighter, NaturallyVulnerableToSplash, Humanoid	Swordsmen are simple melee warriors, but do not underestimate them when they come in numbers.
Wasp	E Flyer	25	8	13	2.5	1	2.5	RangedFighter, Flying, Monster	Wasps are flying monsters eager to destroy every single one of your economic buildings.
Hunterling	E Hunterling	30	10	3.5	5	0.5	10	MeeleFighter, FastMoving, Monster, NaturallyVulnerableToSplash	Hunterlings are incredibly fast and dangerous monsters with the sole aim of chasing you down. They can smell y
Racer	E Racer	35	10	3.5	2.5	1	2.5	MeeleFighter, TakesReducedDamageFromPlayerAttacks, FastMoving, Monster	Racers are fast rolling monsters that are not messing around. They aim right for your castle center and are so
Catapult	E Catapult	45	2	26.4	150	5	30	RangedFighter, SiegeWeapon	Catapults are siege units that deal massive damage to your buildings from afar. Your only chance is to get clo
Ogre	E Ogre	200	4	3	20	1	20	MeeleFighter, NaturallyHighHealthTarget, Humanoid	Ogres are huge melee warriors that really pack a punch. Hey! Due to their human heritage, ogres aren't monster

Damage is the first row of the weapon's damage table (per-victim-tag multipliers apply in *Hp.TakeDamage*; see the combat doc).

5. Buildings, costs and unlocks

Building	Slots	Role	Slot appears when (activator upgrade #, count)	Prerequisite gold spent first (count)	Levels	Gold cost per level	Max income/day	Level-cap diff
Castle Center	1	castle	from the start ×1	0 g ×1	4	3 → 7 → 20 → 100	20	–
Field	15	economy	Mill upgrade #1 ×6, Mill upgrade #2 ×9	6 g ×6, 10 g ×9	1	1	1	–
House	11	economy	Castle Center upgrade #1 ×1, House upgrade #1 ×6, Wall upgrade #1 ×1, Castle Center upgrade #2 ×2, Castle Center upgrade #3 ×1	3 g ×1, 5 g ×1, 7 g ×1, 9 g ×1, 10 g ×2, 11 g ×1, 12 g ×1, 14 g ×1, 16 g ×1, 30 g ×1	3	2 → 2 → 20	4	0
Mill	3	economy	Castle Center upgrade #1 ×3	3 g ×3	3	3 → 4 → 6	6	0
Archery Range	2	military	Castle Center upgrade #2 ×1, Castle Center upgrade #3 ×1	10 g ×1, 30 g ×1	4	4 → 8 → 16 → 50	0	0
Barracks	2	military	Castle Center upgrade #2 ×1, Castle Center upgrade #3 ×1	10 g ×1, 30 g ×1	4	4 → 8 → 16 → 50	0	0
Summoning Circle	2	other	Castle Center upgrade #1 ×2	3 g ×2	2	1 → 2	0	0
Shrine	6	shrine	Castle Center upgrade #1 ×6	3 g ×6	1	3	0	0
Blacksmith	1	support	Castle Center upgrade #1 ×1	3 g ×1	3	4 → 9 → 16	0	0
Defense Tower	10	tower	Castle Center upgrade #1 ×2, Castle Center upgrade #2 ×4, Castle Center upgrade #3 ×4	3 g ×2, 10 g ×4, 30 g ×4	4	3 → 5 → 15 → 40	3	0

Building	Slots	Role	Slot appears when (activator upgrade #, count)	Prerequisite gold spent first (count)	Levels	Gold cost per level	Max income/day	Level-cap diff
Wall	16	wall	Wall upgrade #1 × 12, Castle Center upgrade #2 × 1, Castle Center upgrade #3 × 3 (shares its activator's upgrades)	10 g × 1, 14 g × 6, 30 g × 3, 33 g × 6	3	3 → 2 → 50	0	1

'upgrade #N' means the slot appears when its activator is bought for the N-th time (activator level $N - 1 \rightarrow N$; `BuildSlot.Activate`: `activatorBuilding.Level > activatorLevel`). 'Prerequisite gold' sums those activator upgrades plus, recursively, what the activator itself needed. Level cap: past level 0 a building can only be upgraded while its root building's level is greater than its own level + the diff (`BuiLdSlot.CanBeUpgraded`); with diff 0 a Barracks can never exceed the castle's level.

Unlock timeline (derived): 0 g → 1× Castle Center; **3 g** → 1× Blacksmith, 2× Defense Tower, 1× House, 3× Mill, 6× Shrine, 2× Summoning Circle; **5 g** → 1× House; **6 g** → 6× Field; **7 g** → 1× House; **9 g** → 1× House; **10 g** → 1× Archery Range, 1× Barracks, 4× Defense Tower, 9× Field, 2× House, 1× Wall; **11 g** → 1× House; **12 g** → 1× House; **14 g** → 1× House, 6× Wall; **16 g** → 1× House; **30 g** → 1× Archery Range, 1× Barracks, 4× Defense Tower, 1× House, 3× Wall; **33 g** → 6× Wall.

Totals: building every slot to its final level costs **2348 gold**; if everything is at maximum level the buildings pay **127 gold per dawn** (`goldIncomeChange`, paid at dawn only — see the economy doc).

Castle Center

Lvl	Gold	Cores	Choice	Income	HP +	Tooltip	Activates
1	3					Start a new Kingdom!	Lvl 1 Perks
2	7		Royal Training		200	Unlock more build options and choose an upgrade.	Royal Training; Auto Attack Level 2 [range 35, dps 2.25]; Royal Training Mesh
2	7		Builder's Guild		200	Unlock more build options and choose an upgrade.	Builder's Guild; Auto Attack Level 2 [range 35, dps 2.25]; Builders Guild Mesh
2	7		Magic Armor		200	Unlock more build options and choose an upgrade.	Magic Armor; Auto Attack Level 2 [range 35, dps 2.25]; Magic Armor Mesh
2	7		Assassin's Training		200	Unlock more build options and choose an upgrade.	Assassin's Training; Auto Attack Level 2 [range 35, dps 2.25]; Assassins Training Mesh
3	20		Royal Mastery		400	Unlock more build options and choose an upgrade.	Auto Attack Level 3 [range 35, dps 4.5]; Royal Mastery; Royal Mastery Mesh
3	20		Commander		400	Unlock more build options and choose an upgrade.	Commander; Auto Attack Level 3 [range 35, dps 4.5]; Commander Mesh
3	20		Castle Up		400	Unlock more build options and choose an upgrade.	Castle Up; Auto Attack Level 3 [range 35, dps 4.5]; Castle Up Mesh
3	20		Godly Curse		400	Unlock more build options and choose an upgrade.	Godly Curse; Auto Attack Level 3 [range 35, dps 4.5]; Godly Curse Mesh
4	100		Empire Towers		1000		Empire Towers
4	100		Empire Troups		1000		Empire Troups
4	100		Empire Armor		1000		Empire Armor
4	100		Empire Growth	20	1000		Empire Growth

Defense Tower

Lvl	Gold	Cores	Choice	Income	HP +	Tooltip	Activates
1	3					Shoots at enemies.	Tower Attack [range 36, dps 4]
2	5		Castle Tower		150	Improves health and attack damage.	Tower Attack [range 36, dps 4]; Castle Tower
2	5		Sniper Tower			Improves health and attack damage.	Tower Attack [range 36, dps 4]; Sniper Tower
2	5		Armored Tower		350	Improves health and attack damage.	Tower Attack [range 36, dps 4]; Armored Tower
2	5		Bunker Tower		75	Improves health and attack damage.	Tower Attack [range 36, dps 4]; Bunker Tower
3	15		Archers Spire		350	Choose between several upgrades.	Tower Attack [range 36, dps 4]; Archers Spire; Archers Spire Visuals
3	15		Ballistic Spire			Choose between several upgrades.	Tower Attack [range 36, dps 4]; Ballistic Spire; Ballistic Spire Visuals
3	15		Fire Spire		250	Choose between several upgrades.	Tower Attack [range 36, dps 4]; PourHotOil; Fire Spire; Fire Spire Visuals

Lvl	Gold	Cores	Choice	Income	HP +	Tooltip	Activates
3	15		Healing Spire		750	Choose between several upgrades.	Tower Attack [range 36, dps 4]; Healing Spire Visuals; Healing Spire
4	40		Insane Damage			Choose between several upgrades.	Tower Attack [range 36, dps 4]; Insane Damage
4	40		Insane Range			Choose between several upgrades.	Tower Attack [range 36, dps 4]; Insane Range
4	40		Insane Health		2500	Choose between several upgrades.	Tower Attack [range 36, dps 4]; Insane Health
4	40		Hunting Equipment	3		Choose between several upgrades.	Tower Attack [range 36, dps 4]; Hunting Equipment

Barracks

Lvl	Gold	Cores	Choice	Income	HP +	Tooltip	Activates
1	4		Knights			Choose a melee unit type.	Knights [12× Knights (HP 105, speed 4, range 3.5, dps 2.5); respawn s by level [15.0, 10.0, 6.0, 6.0]]; 4× Knights [HP 105; range 3.5, dps 2.5]
1	4		Speermen			Choose a melee unit type.	Speers [12× Spearmen (HP 66, speed 9.5, range 5.5, dps 4.04); respawn s by level [15.0, 10.0, 6.0, 6.0]]; 4× Spearmen [HP 66; range 5.5, dps 4.04]
1	4		Flails			Choose a melee unit type.	Morningstars [12× Flails (HP 70, speed 5, range 4, dps 1.65); respawn s by level [15.0, 10.0, 6.0, 6.0]]; 4× Flails [HP 70; range 4, dps 1.65]
1	4		Berserks			Choose a melee unit type.	Berserks [12× Berserks (HP 58, speed 6.5, range 3, dps 5); respawn s by level [15.0, 10.0, 6.0, 6.0]]; 4× Berserks [HP 58; range 3, dps 5]
2	8				10	More units, faster unit respawn.	4× Knights [HP 105; range 3.5, dps 2.5]; 4× Spearmen [HP 66; range 5.5, dps 4.04]; 4× Flails [HP 70; range 4, dps 1.65]; 4× Berserks [HP 58; range 3, dps 5]
3	16				10	More units, faster unit respawn.	4× Knights [HP 105; range 3.5, dps 2.5]; 4× Spearmen [HP 66; range 5.5, dps 4.04]; 4× Flails [HP 70; range 4, dps 1.65]; 4× Berserks [HP 58; range 3, dps 5]
4	50				150	Stronger Units	Upgrade To Lvl 4

Archery Range

Lvl	Gold	Cores	Choice	Income	HP +	Tooltip	Activates
1	4		Longbow Archers			Choose a ranged unit type.	Longbows [12× Longbow Archers (HP 25, speed 9, range 50, dps 2); respawn s by level [15.0, 10.0, 6.0, 6.0]]
1	4		Crossbowpeople			Choose a ranged unit type.	Crossbows [12× Crossbowmen (HP 45, speed 4, range 21, dps 3.5); respawn s by level [15.0, 10.0, 6.0, 6.0]]
1	4		Hunters			Choose a ranged unit type.	Hunters [12× Hunters (HP 45, speed 6, range 35, dps 1.96); respawn s by level [15.0, 10.0, 6.0, 6.0]]
1	4		Fire Archers			Choose a ranged unit type.	Fire Archers [12× Fire Archers (HP 20, speed 3.5, range 25, dps 1.5); respawn s by level [15.0, 10.0, 6.0, 6.0]]
2	8				10	More units, faster respawn.	4× Longbow Archers [HP 25; range 50, dps 2]; 4× Crossbowmen [HP 45; range 21, dps 3.5]; 4× Hunters [HP 45; range 35, dps 1.96; range 3.5, dps 5.48]; 4× Fire Archers [HP 20; range 25, dps 1.5]
3	16				10	More units, faster respawn.	4× Longbow Archers [HP 25; range 50, dps 2]; 4× Crossbowmen [HP 45; range 21, dps 3.5]; 4× Hunters [HP 45; range 35, dps 1.96; range 3.5, dps 5.48]; 4× Fire Archers [HP 20; range 25, dps 1.5]
4	50				150	More units, faster respawn.	Upgrade To Lvl 4

Blacksmith

Lvl	Gold	Cores	Choice	Income	HP +	Tooltip	Activates
1	4		Melee Attack			Can research upgrades. Upgrades apply to you, your troup	MeleeAttack; MA1
1	4		Ranged Attack			Can research upgrades. Upgrades apply to you, your troup	RangedAttack; RA1
1	4		Melee Armor			Can research upgrades. Upgrades apply to you, your troup	MeleeResistance; 2× MR1

Lvl	Gold	Cores	Choice	Income	HP +	Tooltip	Activates
1	4		Ranged Armor			Can research upgrades. Upgrades apply to you, your troupes and	RangedResistance; 2× RR1
2	9		Melee Attack			Researches upgrades. Upgrades apply to you, your troupes and	MeleeAttack; MA2
2	9		Ranged Attack			Researches upgrades. Upgrades apply to you, your troupes and	RangedAttack; RA2
2	9		Melee Armor			Researches upgrades. Upgrades apply to you, your troupes and	MeleeResistance; 2× MR2
2	9		Ranged Armor			Researches upgrades. Upgrades apply to you, your troupes and	RangedResistance; 2× RR2
3	16		Melee Attack			Researches upgrades. Upgrades apply to you, your troupes and	MeleeAttack; MA3
3	16		Ranged Attack			Researches upgrades. Upgrades apply to you, your troupes and	RangedAttack; RA3
3	16		Melee Armor			Researches upgrades. Upgrades apply to you, your troupes and	MeleeResistance; 2× MR3
3	16		Ranged Armor			Researches upgrades. Upgrades apply to you, your troupes and	RangedResistance; 2× RR3

Mill

Lvl	Gold	Cores	Choice	Income	HP +	Tooltip	Activates
1	3		Improved Plow	1		Unlocks the option to build fields to increase income.	Auto Attack Lvl 1 [range 38, dps 1.21]; Lvl 1 Rotor; Improved Plow
1	3		Explosive Trap	1		Unlocks the option to build fields to increase income.	Auto Attack Lvl 1 [range 38, dps 1.21]; Lvl 1 Rotor; Explosive Trap
1	3		Scarecrows	1		Unlocks the option to build fields to increase income.	Auto Attack Lvl 1 [range 38, dps 1.21]; Lvl 1 Rotor; Scarecrows
1	3		Wind Spirits	1		Unlocks the option to build fields to increase income.	Auto Attack Lvl 1 [range 38, dps 1.21]; Lvl 1 Rotor; Wind Spirits [Auto Attack Lvl 1: range 38, dps 1.21; Auto Attack Lvl 2: range 38, dps 2; Auto Attack Lvl 3: range 38, dps 4]
2	4			1	20	Increases income significantly.	Lvl 2 Rotor; Auto Attack Lvl 2 [range 38, dps 2]
3	6			4	30	Increases income significantly.	Lvl 3 Rotor; Auto Attack Lvl 3 [range 38, dps 4]

House

Lvl	Gold	Cores	Choice	Income	HP +	Tooltip	Activates
1	2			1		A stable income source.	AutoAttackPerk Lvl1 [range 34, dps 1.6]
2	2			1	10	Increases daily income.	AutoAttackPerk Lvl2 [range 34, dps 4.82]
3	20			2	70	Increases daily income.	

Shrine

Lvl	Gold	Cores	Choice	Income	HP +	Tooltip	Activates
1	3					Shoots at enemies.	

6. Practical numbers for planning

- Hero-reachable ground: **7822 m²** (unbuilt world) of 8087 m² navmesh; 106 blocking colliders ({'static': 11, 'building': 95}), 12 auto-opening gates.
- The hero can walk from the castle to the nearest spawn-line vertex in 47–86 m (2.6 s at day speed).

- 69/69 build slots have at least one collision-free stand point within interaction range (`terrain/05_Nordfels.json` → `slots[].standPoints`); median walking distance castle→stand point 30 m.
- 6 shrines, each needs **350 XP**, collection range 20 m, pays 2 gold/dawn once unlocked (XP comes from unit deaths inside the range).

7. Method notes and limits

- **Sources:** waves, spawn lines, slots, unit and enemy stats are `MonoBehaviour` payloads decoded with layouts generated from the game DLLs (byte-exact on 655,330/655,330 objects); navmesh from the baked A* cache; colliders from the scene's physics objects. See `tools/refpack/`.
- **Hero-walkable model:** navmesh surface connected to the castle, minus colliders (blockers evaluated at the local surface height, walkable ramps excluded by triangle normal, gates ignored). It contains the hero in 99.7 % of logged Nordfels ticks (telemetry validation).
- **Estimates:** arrival times ignore fighting and avoidance; income is an upper bound; unit dps ignores per-victim multipliers and upgrades from perks/blacksmith.